Xihan Ji - Curriculum Vitæ

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EMPLOYMENT

Zhejiang University, *Institute for Astronomy* — *Hangzhou, China*
Assistant Professor — Since Oct 2026

University of Cambridge, *Cavendish Laboratory & KICC* — *Cambridge, UK*
Postdoctoral Research Associate — 2023 - 2026

EDUCATION

University of Kentucky, *Department of Physics and Astronomy* — *Lexington, US*
Ph.D. in Physics; Supervisors: Profs. Renbin Yan and Gary Ferland — 2023
Dissertation: "A Multidimensional View on the Emission-line Diagnostics of the Warm Ionized Gas in Nearby Galaxies"

Tsinghua University, *Department of Physics* — *Beijing, China*
B.Sc. in Physics; Supervisor: Prof. Cheng Li — 2018

RESEARCH EXPERIENCE & SKILLS

I work on observational data from state-of-the-art telescopes including *JWST*, VLT, Gemini, and GTC. Many of my works are also based on the photoionization code Cloudy.

I have been working on the following research topics.

Galaxy formation and evolution at high redshift with NIR spectroscopy

· Formation and evolution of supermassive black holes with *JWST*.
· Chemical enrichment of galaxies with *JWST*.
· Local analogs of high-redshift active galactic nuclei (AGN) with ground-based telescopes.

Physics of the warm ionized gas in the local Universe with optical IFS

· Calibrations of metallicities, ionization parameters, and dust attenuation curves in nearby ($z \sim 0.03$) galaxies with SDSS-IV/MaNGA.
· Testing photoionization models with Galactic H II regions.

COLLABORATIONS

I am a team member of the following collaborations.

- **The JWST Advanced Deep Extragalactic Survey (JADES)**
- **The JWST BlackTHUNDER Large Program**
- **The VLT MOONRISE GTO Survey** - AGN & metallicity working groups
- **The Affordable Multiple Aperture Spectroscopy Explorer (AMASE)** - Metallicity and ionization model working group (Group Lead)
- **The Dark Energy Spectroscopic Instrument (DESI) survey**
- **The SDSS-IV/MaNGA survey** - Data analysis team (Core Member)

FELLOWSHIPS & AWARDS

Presidential Fellowship, University of Kentucky 2022

SELECTED TALKS

Seminars & Lectures

Astro Seminar, Imperial College London, UK **(invited)**	Apr, 2026
Astro Colloquium, Durham University, UK **(invited)**	Oct, 2025
Astro Seminar, MPI for Gravitational Physics, Germany **(invited)**	Nov, 2024
International Max Planck Research School Lecture Week, Germany **(invited)**	Sep, 2024
Astro Seminar, The Chinese University of Hong Kong, China **(invited)**	Aug, 2023
Astro Seminar, Nanjing University, China **(invited)**	June, 2023
Star Formation/ISM Rendezvous, Princeton University, US	Dec, 2022
Astro Lunch, University of Pittsburgh, US **(invited)**	Jan, 2021

Conferences & Workshops

Infant Black Holes and Early Galaxies, Rome, Italy **(invited)**	Oct, 2026
Little Red Dot Workshop, Online **(invited)**	June, 2026
GalFirenze, Florence, Italy	June, 2026
Charting Cosmic Dawn in Copenhagen, Helsingør, Denmark	Apr, 2026
Unraveling Little Red Dots, Trieste, Italy **(invited)**	Oct, 2025
Massive Black Holes across Cosmic Time, Cambridge, UK	Sep, 2025
Big Galaxies, Big Problems, Leiden, Netherlands **(invited)**	Apr, 2025
First Galaxies: Building blocks of galaxies across cosmic time, Oxford, UK	Apr, 2025
The 40th IAP Symposium, Paris, France	Dec, 2024
The Origin and Evolution of Supermassive Black Holes, Sexten, Italy **(invited)**	July, 2024
First Stars VII, New York City, US	May, 2024
Resolving Galaxy Ecosystems Across All Scales, Hong Kong, China **(invited)**	Dec, 2023

FUNDING AND RESEARCH INITIATIVE

I led/co-led the following selected accepted programmes (for some of them I am not listed as PI or co-PI due to application policies).

- **Weighing the dust of black-hole stars – deepest probe of the dust continua in low-redshift "Little Red Dots" with ALMA**, ALMA, (17 hours; PI: Ji, X), 2026
- **Probing BLR obscuration in a unique AGN with varying Balmer-line absorption at z=0.29 with Chandra and Gemini**, (25 ks Chandra + 1 hour Gemini; PI: Ji, X), 2026
- **X-ray census of "Little Red Dot" AGN at $z \approx 0.2 - 0.4$ with XMM-Newton**, XMM-Newton (144 ks; PI: Ji, X), 2026.
- **Déraciné: Characterizing a Local "Little Red Dot" with Extreme Line Equivalent Widths at $z = 0.25$ with X-shooter**, VLT/X-Shooter spectrograph (7 hours; PI: Ji, X), 2026.
- **Characterizing the gaseous environments of accreting black holes in 6 "Little Red Dots" at $z = 0.2 - 0.4$ with GTC/ORSIRIS**, GTC/OSIRIS spectrograph (30 hours; PI: Ramos Almeida, C.), 2026.
- **Optical spectroscopy of the nearest "Little Red Dot" at $z = 0.1$**, GTC/OSIRIS spectrograph (director's discretionary time, 2 hours; PI: Ramos Almeida, C.), 2025.

- **Hard or soft SED? A radio probe to the heart of the brightest Little Red Dot**, VLA (8.5 hours; PI: Nelson, E.), 2025.
- **Mapping spatially resolved kinetic temperatures of H II regions**, WIYN/HexPak integral field spectroscopy (3 nights; PIs: Bershady, M. & Yan, R.), 2023.

Total observing time worth approximately €600,000.

TEACHING AND SUPERVISING EXPERIENCE

· Taught a remote, 8-week ad-hoc mini-course on "Photoionization Models and Nebular Diagnostics" (total 16 contact hours with weekly homework) for a group of Master and Ph.D. students at The Chinese University of Hong Kong in 2022.
· Teaching assistant in "General University Physics Laboratory" for undergraduate students at the University of Kentucky in Fall 2019.

· Co-supervision of Lucy Ivey - 2nd-year Ph.D. in Astrophysics
· Co-supervision of Urvi Thakurdesai - 1st-year Ph.D. in Astrophysics
· Co-supervision of Ignas Juodžbalis - Ph.D. in Astrophysics - 2026
· Co-supervision of Greta Zucchi - Master in Astrophysics - 2025
· Co-supervision of Bartolomeo Trefoloni - Ph.D. in Astrophysics - 2025
· Co-supervision of Chamani Gunaksekera - Ph.D. in Physics - 2023

PROFESSIONAL SERVICES & SOCIAL ACTIVITIES

- **Panel/Peer reviews:** Served as a reviewer for NASA Astrophysics Data Analysis Program (ADAP); served as a reviewer for Nature, Nature Astronomy, ApJ, and A&A.
- **Press release:** The Gran Telescopio Canarias observes the closest "Little Red Dot," shedding light on how the first black holes grew.
- **Data release:** Wrote an emission-line fitting module for the Data Analysis Pipeline of the SDSS-IV MaNGA Survey, included in subsequent SDSS public data releases.
- **Organizing academic activities:** Lead organizers of the workshop, "Linking early black holes and stellar evolution with Little Red Dots", in Cambridge, 2026; LOC of the conference, "Massive Black Holes across Cosmic Time", in Cambridge, 2025; organize the weekly extragalactic group meeting in Cambridge since 2025; organized the weekly Astro Journal Club at the University of Kentucky from 2020 to 2022.
- **Outreach:** Demonstration of the grating spectroscopy at the 2024 KICC/IoA Open Day in Cambridge, UK; volunteered in the "Full STEM Ahead: Solar Telescopes" program in Lexington, US.

PUBLICATIONS

My ADS Library